



MODULE NAME:	MODULE CODE:
SYSTEM ANALYSIS AND DESIGN	SAND6211/d/p
SYSTEM ANALYSIS AND DESIGN	SAND6221/d

ASSESSMENT TYPE:	TAKE-HOME ASSESSMENT (PAPER ONLY)
TOTAL MARK ALLOCATION:	60 MARKS
TOTAL TIME:	This assessment should take you 1 Hour to complete, however, you have 21 Hours (midnight to 9PM on the same day) to submit. This additional time has been allocated to allow for the download, completion, and upload of your submission.

By submitting this assessment, you acknowledge that you have read and understood all the rules as per the terms in the registration contract, in particular the assignment and assessment rules in The IIE Assessment Strategy and Policy (IIE009), the intellectual integrity and plagiarism rules in the Intellectual Integrity Policy (IIE023), as well as any rules and regulations published in the student portal.

INSTRUCTIONS:

1. Please **adhere to all instructions**. These instructions are different from what is normally present, so take time to go through these carefully.
2. **Independent work is required**. Students are not allowed to work together on this assessment. Any contraventions of this will be handled as per disciplinary procedures in The IIE policy.
3. **No material may be copied from original sources, even if referenced correctly, unless it is a direct quote indicated with quotation marks.**
4. All work must be adequately and correctly referenced.
5. You should paraphrase (use your own words) the concepts that you are referencing, rather than quoting directly.
6. Marks will be awarded for the quality of your paraphrasing.
7. This is an open-book assessment.
8. Assessments must be typed unless otherwise specified.
9. **Ensure that you save a copy of your responses.**
 - a. Complete your responses in a Word document.
 - b. The document name must be your **name.student number.Module Code**.
 - c. Once you have completed the assessment, upload your document under the **submission link** in the correct module in Learn.

Additional instructions:

- Calculators are allowed
- For multiple-choice questions, give only one (1) response per question. The marker will ignore any question with more than one answer, unless otherwise stated. You should, therefore, be sure of your answer before committing it to paper.
- Answer All Questions. .
- Show all calculations, where applicable (marks may be awarded for this).

Referencing Rubric

Providing evidence based on valid and referenced academic sources is a fundamental educational principle and the cornerstone of high-quality academic work. Hence, The IIE considers it essential to develop the referencing skills of our students in our commitment to achieve high academic standards. Part of achieving these high standards is referencing in a way that is consistent, technically correct and congruent. This is not plagiarism, which is handled differently.

Poor quality formatting in your referencing will result in a penalty of a maximum of ten percent being deducted from the percentage awarded, according to the following guidelines. Please note, however, that **evidence of plagiarism in the form of copied or uncited work (not referenced), absent reference lists, or exceptionally poor referencing, may result in action being taken in accordance with The IIE's Intellectual Integrity Policy (0023).**

Markers are required to provide feedback to students by indicating (circling/underlining) the information that best describes the student's work.

Minor technical referencing errors: 5% deduction from the overall percentage – the student's work contains **five or more errors** listed in the minor errors column in the table below.

Major technical referencing errors: 10% deduction from the overall percentage – the student's work contains **five or more errors** listed in the major errors column in the table below.

If both minor and major errors are indicated, then 10% only (and not 5% or 15%) is deducted from the overall percentage. The examples provided below are not exhaustive but are provided to illustrate the error

Required: Technically correct referencing style	Minor errors in technical correctness of referencing style Deduct 5% from percentage awarded	Major errors in technical correctness of referencing style Deduct 10% from percentage awarded
<u>Consistency</u> The same referencing format has been used for all in-text references and in the bibliography/reference list.	Minor inconsistencies. - The referencing style is generally consistent, but there are one or two changes in the format of in-text referencing and/or in the bibliography. - For example, page numbers for direct quotes (in-text) have been provided for one source, but not in another instance. Two book chapters (bibliography) have been referenced in the bibliography in two different formats.	Major inconsistencies. - Poor and inconsistent referencing style used in-text and/or in the bibliography/ reference list. - Multiple formats for the same type of referencing have been used. - For example, the format for direct quotes (in-text) and/or book chapters (bibliography/ reference list) is different across multiple instances.
<u>Technical correctness</u> - Referencing format is technically correct throughout the submission. - Position of the reference: a reference is directly associated with every concept or idea. - For example, quotation marks, page numbers, years, etc. are applied correctly, sources in the bibliography/reference list are correctly presented.	Generally, technically correct with some minor errors. - The correct referencing format has been consistently used, but there are one or two errors. - Concepts and ideas are typically referenced, but a reference is missing from one small section of the work. - Position of the references: references are only given at the beginning or end of every paragraph. - For example, the student has incorrectly presented direct quotes (in-text) and/or book chapters (bibliography/reference list).	Technically incorrect. - The referencing format is incorrect. - Concepts and ideas are typically referenced, but a reference is missing from small sections of the work. - Position of the references: references are only given at the beginning or end of large sections of work. - For example, incorrect author information is provided, no year of publication is provided, quotation marks and/or page numbers for direct quotes missing, page numbers are provided for paraphrased material, the incorrect punctuation is used (in-text); the bibliography/reference list is not in alphabetical order, the incorrect format for a book chapter/journal article is used, information is missing e.g. no place of publication had been provided (bibliography); repeated sources on the reference list.
<u>Congruence between in-text referencing and bibliography/ reference list</u> All sources are accurately reflected and are all accurately included in the bibliography/ reference list.	Generally, congruence between the in-text referencing and the bibliography/ reference list with one or two errors. - There is largely a match between the sources presented in-text and the bibliography. - For example, a source appears in the text, but not in the bibliography/ reference list or vice versa.	A lack of congruence between the in-text referencing and the bibliography. - No relationship/several incongruities between the in-text referencing and the bibliography/reference list. - For example, sources are included in-text, but not in the bibliography and vice versa, a link, rather than the actual reference is provided in the bibliography.
In summary: the recording of references is accurate and complete.	In summary, at least 80% of the sources are correctly reflected and included in a reference list.	In summary, at least 60% of the sources are incorrectly reflected and/or not included in reference list.

Overall Feedback about the consistency, technical correctness and congruence between in-text referencing and bibliography:

Question 1	(Marks: 20)
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Answer all of the questions below.

Q.1.1	By using your own words, define a Subsystem and briefly discuss the importance of dividing an information system into subsystems. Provide a real-life example of a system with one or more subsystems. Please use your own words.	(6)
Q.1.2	Briefly explain the purpose of SDLC and discuss the importance of the first two core processes of the SDLC. Please use your own words.	(6)
Q.1.3	Briefly explain what stakeholders are in system development and provide two examples.	(4)
Q.1.4	There are different types of events to consider when using the Event Decomposition Technique. Define what the Event Decomposition Technique is and distinguish between external and state events.	(4)

Question 2	(Marks: 20)
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The questions below relate to the following case study:

System: Technical Help Desk System

Dawn Help Desk Corporation is a global help desk company that works with leading multi-national technology corporations. The company offers help desk consultants who assists multi-national technology corporations with their technical help desk, technical support and after sales consultations. Due to the rapid growth, a Software Development department was established to improve and automate all the manual processes and workflows. The department is currently conducting a study in order to develop a new Technical Help Desk System to assist the help desk team with data capturing and information management.

The help desk team currently have an established and strict manual business process and workflow that they follow whenever they assist a customer. A help desk consultant collects customer details, and registers an incident, and sends it to a Technical Consultant for actioning. The Technical Consultant receives an incident, conducts initial troubleshooting, compiles an initial technical report and sends it to the Supervisor. Once finalized the initial technical report is forwarded to Supervisor. The Supervisor receives the initial technical report, confirms the details, and approves the report.

As part of the business rules, a customer will be allowed to log more than one incident for items to be fixed. Using the serial number of the purchased item, the customer would be allowed to log incidents for many items. The maintenance account is allowed to idle with no item listing. Items belong to certain categories or are categorized. All Help desk consultants, Technical Consultants and Supervisors are all permanent employees of Dawn Help Desk Corporation and their information will be stored in one table on the database. The Help Desk Consultant would be assigned to many customers maintenance accounts. Lastly the maintenance account is allowed to idle with no Help desk consultant assigned to it.

In terms of technology, the system will consist of a relational database based on a Microsoft SQL Server relational database management system (RDBMS) and will be developed using ASP.Net Core and other open-source platforms. The Software Development department at Dawn Help Desk Corporation has adopted agile project management instead of relying on the traditional approach to project management. The developers will handle their own Unit testing and further testing on the system will be outsourced to one of the leading Testing Consulting Company in the world because of the standards imposed by the multi-national technology corporations that have entrusted Dawn Help desk Corporation with their help desk operations.

Q.2.1	Based on Dawn Help desk Corporation's current established and strict manual business processes and workflow outlined in the case study, develop a comprehensive UML activity diagram to model the workflows.	(20)
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Question 3		(Marks: 20)
The questions below are still based on the Technical Help Desk System case study in Question 2.		
Q.3.1	As stated in the case study, all the databases on Postgres including the back-ups should be encrypted. Discuss the importance of encryption, and distinguish between encryption and decryption in computer security.	(6)
Q.3.2	The case study has numerous use cases and detailed information about use case is described with a use case description. List any four aspects of a use case covered in a use case description.	(4)
Q.3.3	In today's interconnected world, systems need reliable access control systems to keep the data secure. List and define the three elements that access control systems rely on.	(6)
Q.3.4	Discuss two things you would take into consideration when designing the interface for both Web and Mobile.	(4)

END OF PAPER